

SKILL DEVELOPMENT REPORT

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

(Smart Agriculture and security System)

(2JR24EC021)

DLithe Consultancy Services Pvt. Ltd.



Skill Development Report

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Job Assignment: IoT skill development project

Project name : Smart Agriculture System

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Observations:

Submitted to

Signature of Trainer

Date:

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Date:

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1. INTRODUCTION

Smart irrigation systems are modern agricultural solutions that use sensors and automation technology to provide water to crops efficiently. In traditional farming methods, irrigation is done manually, which may lead to water wastage, excessive labour, and improper crop monitoring. To overcome these problems, an IoT-based Smart Irrigation and Security System is developed using the ESP32 microcontroller and different sensors.

In this project, components such as a soil moisture sensor, DHT11 temperature and humidity sensor, ultrasonic sensor, buzzer, LCD display, water pump, and L298N motor driver are connected using a breadboard for easy interfacing and testing. The ESP32 acts as the main controller of the system and processes data collected from the sensors.

The soil moisture sensor continuously checks the moisture level of the soil. If the soil becomes dry, the ESP32 automatically turns ON the water pump through the L298N motor driver. When the soil contains sufficient moisture, the motor turns OFF automatically, helping to save water and improve irrigation efficiency.

The DHT11 sensor monitors environmental temperature and humidity conditions, while the ultrasonic sensor is used to detect animals or obstacles near the agricultural field. When an object is detected within a certain distance, the buzzer gives an alert for farm protection. The LCD display shows real-time sensor values and system status.

This smart irrigation and security system reduces manual work, conserves water, improves crop monitoring, and supports modern smart farming techniques using IoT technology.

2. PROBLEM STATEMENT

Agriculture is one of the most important sectors for food production and economic development. However, traditional farming methods face many challenges such as inefficient water usage, lack of continuous monitoring, and poor field security. Farmers usually irrigate crops manually without knowing the exact moisture condition of the soil, which leads to overwatering or underwatering. This not only wastes water but also affects crop growth and productivity.

Another major problem is the absence of proper monitoring of environmental conditions like temperature, humidity, and soil moisture. Farmers need to visit fields regularly to check crop conditions, which requires more time and labour. Sudden environmental changes may damage crops if immediate action is not taken.

Security is also a serious concern in agricultural fields. Crops are often damaged by animals or intruders, especially during nighttime. Traditional security methods such as fencing and manual supervision are not always effective.

To overcome these problems, an automated Smart Agriculture Irrigation and Security System using IoT technology is developed. The system automatically controls irrigation based on soil moisture levels, continuously monitors environmental conditions, and provides security alerts for unauthorized movement or animal intrusion.

The proposed system helps in reducing water wastage, improving crop productivity, minimizing human effort, and enhancing farm security through automation and smart monitoring.

3. OBJECTIVES OF THE PROJECT

The main objectives of the project are:

- To automate the irrigation system using soil moisture sensing
- To reduce water wastage in agriculture
- To monitor temperature and humidity conditions
- To provide security monitoring using ultrasonic sensor and buzzer
- To reduce manual labor in farming
- To display real-time sensor data on LCD
- To improve agricultural productivity using IoT technology

4. SCOPE OF THE PROJECT

The scope of the Smart Agriculture Irrigation and Security System includes:

- Smart farming application
- Automatic irrigation control
- Water conservation systems
- Remote agricultural monitoring
- Farm protection and security
- Greenhouse automation
- Future enhancement using cloud computing and AI technologies
- Mobile-based monitoring and control
- Crop health monitoring
- Low-cost and user-friendly farming solution

5. METHODOLOGY

The working methodology of the project is divided into step:

Step 1 – ESP32 Microcontroller

- The ESP32 microcontroller acts as the main control unit of the system.
- It reads data from all sensors and controls the output devices automatically.
- The ESP32 continuously monitors the system conditions.

Step 2 – Soil Moisture Sensor

- The soil moisture sensor measures the moisture level of the soil.
- If the soil becomes dry, the sensor sends data to the ESP32.
- The ESP32 decides whether irrigation is required or not.

Step 3 – Motor Driver and Water Pump

- The ESP32 sends a signal to the L298N motor driver module.
- The motor driver activates the DC water pump automatically.
- The pump supplies water to the agricultural field.

Step 4 – Automatic Pump OFF

- The soil moisture sensor continuously checks the soil condition during irrigation.
- When the required moisture level is reached, the ESP32 stops the pump.
- This prevents overwatering and saves water.

Step 5 – DHT11 Sensor

- The DHT11 sensor measures temperature and humidity conditions.
- The sensor sends environmental data to the ESP32 continuously.
- These readings help in monitoring crop conditions.

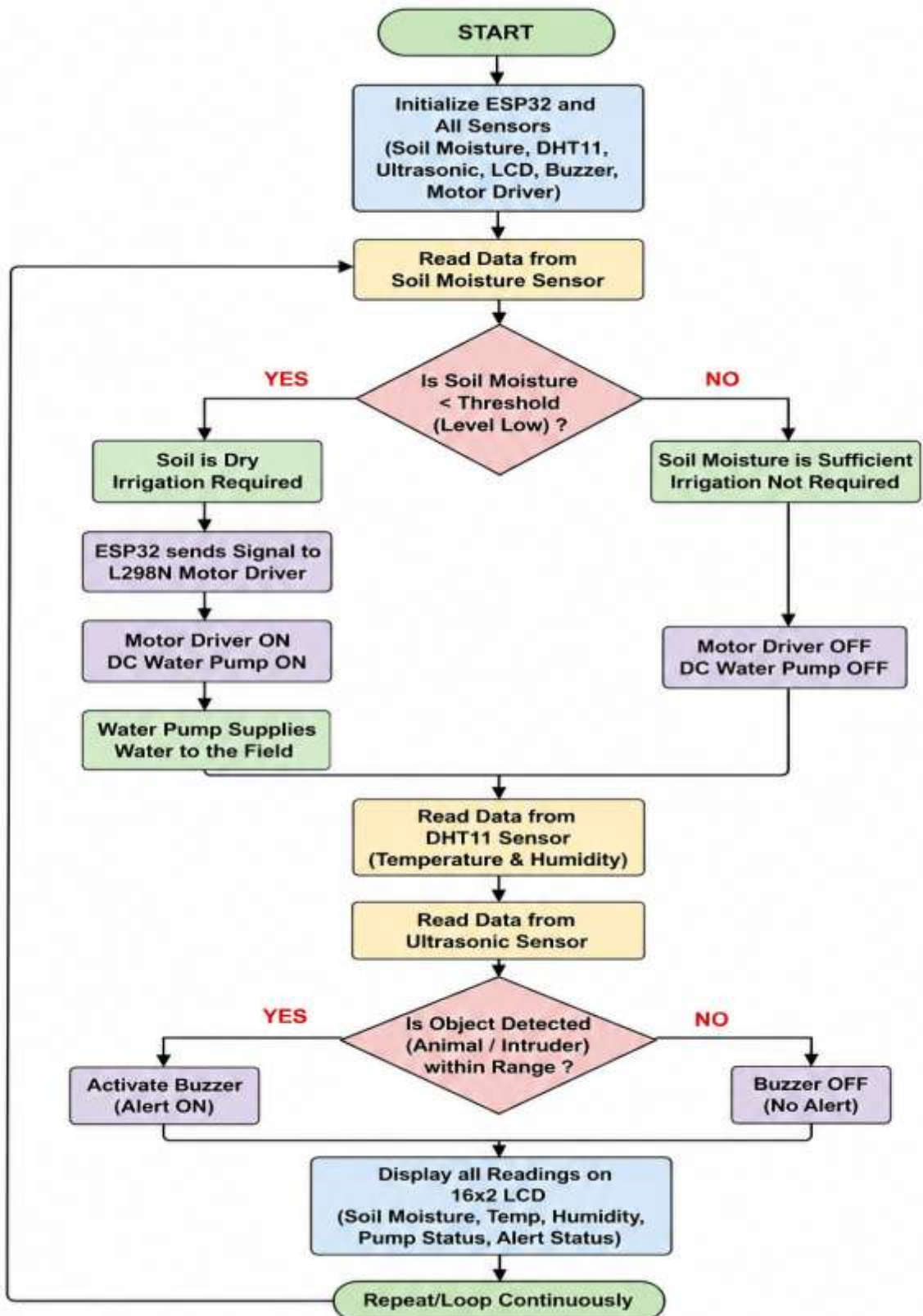
Step 6 – Ultrasonic Sensor and Buzzer

- The ultrasonic sensor detects animals or intruders near the field.
- If any object is detected, the ESP32 activates the buzzer.
- The buzzer gives an alert sound for field security.

Step 7 – LCD Display

- The 16×2 LCD display shows soil moisture, temperature, humidity, and pump status.
- It helps in easy monitoring of the system.

6.FLOW CHART



7. USE CASE DIAGRAM

Actors:

- Admin / Farmer
- User
- Blynk IoT Application
- Sensor System
- Irrigation System
- Security System

Use Cases:

1. Monitor soil moisture level
2. Detect dry soil condition
3. Start automatic irrigation
4. Stop water pump automatically
5. Monitor temperature and humidity
6. Detect animals or intruders
7. Activate buzzer alert
8. Display sensor readings on LCD
9. Monitor pump status
10. Continuous smart field monitoring
11. Water conservation management
12. Provide farm security alerts

8. TOOLS AND TECHNOLOGIES USED

Hardware Components:

Component Name	Purpose
ESP32 Microcontroller	Controls the complete system and processes sensor data
Soil Moisture Sensor	Measures moisture level of the soil
DHT11 Sensor	Measures temperature and humidity
Ultrasonic Sensor	Detects animals or intruders near the field
L298N Motor Driver	Controls the DC water pump
DC Water Pump	Supplies water for irrigation
16x2 LCD Display	Displays sensor readings and system status
Buzzer	Gives alert sound for security warning
Blynk IoT Application	Monitors and controls the system remotely
Jumper Wires & Breadboard	Provides circuit connections
Power Supply / Battery	Supplies power to the system

Software tools:

Software

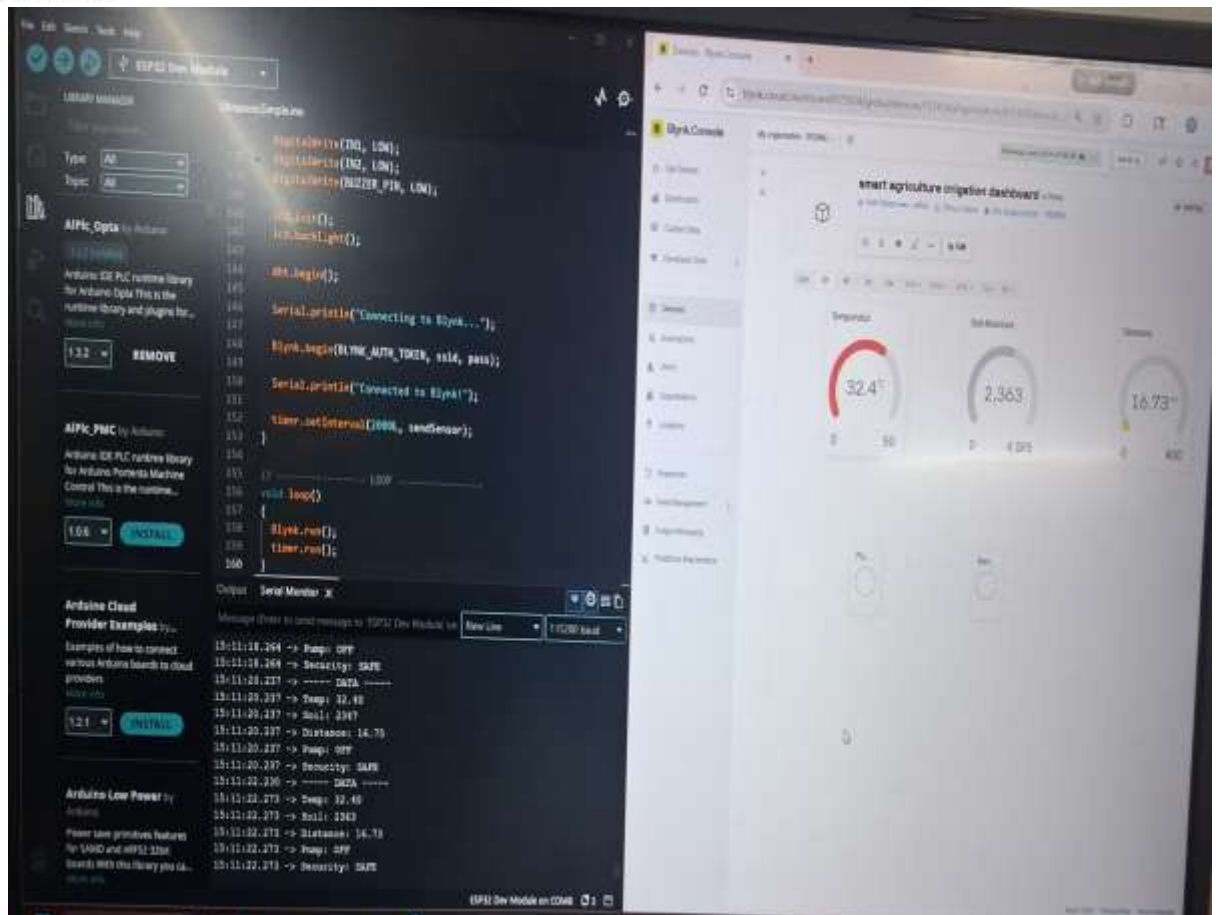
Software Tool	Purpose
Arduino IDE	Used for writing and uploading code to ESP32
Blynk IoT Application	Used for remote monitoring and control
Embedded C Programming	Used for programming the microcontroller
Serial Monitor	Used for debugging and checking sensor values
Wi-Fi Technology	Used for wireless communication and IoT connectivity

9. RESULTS AND SNAPSHOTS

Results

- Automatic irrigation system works successfully.
- Water wastage is reduced significantly.
- Soil moisture monitoring provides accurate irrigation control.
- Temperature and humidity monitoring works properly.
- Security alert system detects nearby objects effectively.
- Buzzer alert activates during animal or intruder detection.
- LCD display shows live system status and sensor values.
- Remote monitoring through the Blynk IoT application works successfully.
- Automatic ON/OFF control of the water pump operates correctly.
- Human effort and manual monitoring are reduced.
- The system provides continuous smart field monitoring.
- Power consumption is minimized through automation.
- Real-time sensor data improves farming decisions.
- Crop protection and farm security are improved.
- Overall farming efficiency and productivity are increased.





10. APPLICATIONS

- Smart agriculture farms
- Greenhouse irrigation systems
- Garden watering systems
- Water conservation projects
- Farm security monitoring
- Automated agricultural systems
- Smart irrigation management
- Crop monitoring systems
- Remote farm monitoring using IoT
- Agricultural research projects
- Home gardening automation
- Nursery plant monitoring systems
- Smart water management systems
- Solar-powered irrigation systems
- Environmental monitoring systems
- Animal intrusion detection systems

11.MIND MAP



12 .GITHUB REPOSITORY LINK

github.com/dhruvmane2/smart-agriculture-and-security-system

13. CONCLUSION

The Smart Agriculture Irrigation and Security System is an efficient and cost-effective solution for modern farming. The system automates irrigation based on soil moisture conditions, reducing water wastage and human effort. It also monitors environmental conditions and provides security alerts using sensors and buzzer systems.

By using IoT technology and ESP32, the project improves agricultural productivity, supports smart farming, and provides better crop management. This system can play an important role in future agriculture and sustainable farming practices.